METHOD FOR A CRYPTOGRAPHIC SIDE-CHANNEL ATTACK USING ACOUSTIC ANALYSIS OF KEYSTROKE METADATA

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Abstract

A method for a cryptographic side-channel attack that assists in the decryption of a target's encrypted communications. The method uses a covert audio surveillance device, concealed within a piece of office equipment such as an Uninterruptible Power Supply (UPS), to capture the acoustic emanations of a user typing on a computer keyboard. The captured audio is processed to extract metadata, specifically the length and rhythm of keystroke sequences, rather than the specific characters typed. This metadata, such as the exact number of characters in a password, is then used to significantly reduce the search space and computational cost of a brute-force decryption attack against the target's encrypted data streams.

Background of the Invention

Acoustic cryptanalysis, or analyzing the sound of a device to infer sensitive information, is a known field of study. Prior art has focused primarily on attempting to identify individual characters by distinguishing the unique sound of each key being pressed. These methods are often unreliable, require extensive training on a specific keyboard, and can be defeated by ambient noise.

In modern cryptography, a common attack vector against encrypted data is a brute-force attack, where an adversary attempts to guess the encryption key by trying all possible combinations. The computational cost of such an attack is directly related to the length of the key and the length of the data being encrypted. Any information that can reduce the number of possibilities, or "constrain the search space," can make a previously infeasible brute-force attack practical. There is a need for a reliable side-channel attack that can provide metadata about an encrypted plaintext to assist in this process.

Summary of the Invention

The present invention is a method for a side-channel attack that provides crucial metadata for a brute-force decryption effort by analyzing the sound of a target's typing.

The method begins with the placement of a covert audio recording device ("bug") in the target's vicinity. The preferred embodiment is to conceal the bug inside a piece of equipment that has its own power source and is likely to be placed near the computer, such as an Uninterruptible Power Supply (UPS). The bug's sole function is to record the sound of the user's keystrokes.

The captured audio is transmitted to a processing system. The system does not attempt to identify individual characters. Instead, it analyzes the acoustic signal to determine metadata about the typing, primarily:

1. **Sequence Length:** The number of distinct keystroke sounds between pauses, which directly corresponds to the length (number of characters) of a password, username, or command.

2. **Typing Rhythm:** The cadence and timing of the keystrokes, which can provide additional heuristic data.

This extracted metadata is then fed into a brute-force decryption system that is attempting to crack the target's encrypted communications. By knowing the exact length of

the password or other plaintext data corresponding to an encrypted data packet, the decryption system can eliminate a vast number of incorrect guesses. For example, if it is known that a password is exactly 12 characters long, the system can discard all key combinations that would decrypt the packet to a plaintext of any other length. This dramatic reduction in the search space makes the brute-force attack significantly more efficient and more likely to succeed in a practical amount of time.

Brief Description of the Drawings

FIG. 1 is a flowchart illustrating the overall process of the side-channel attack, from acoustic capture to use in decryption.

FIG. 2 is a conceptual diagram showing the placement of the audio bug and the type of metadata being extracted.

Detailed Description of the Invention

Referring to the flowchart in **FIG. 1**, the method begins with a covert audio surveillance device capturing the acoustic emanations of a target's keystrokes (101). A preferred placement for this device is within an Uninterruptible Power Supply (UPS) near the target's computer, as it provides power and proximity.

The captured audio is transmitted to a processing system where it undergoes acoustic analysis (102). This analysis is not designed to identify individual characters, but rather to extract metadata, such as the number of keystrokes in a sequence.

The extracted metadata (103), for example, "Password length = 12 characters," is then provided to a Brute-Force Decryption System (104). This system is simultaneously attempting to decrypt an encrypted data packet (105) captured from the target's network traffic.

The decryption system uses the metadata to constrain its search space (106). Instead of testing all possible keys, it only tests keys that, when applied to the encrypted packet, would result in a plaintext of the length specified by the acoustic metadata. This significantly reduces the number of keys to be tested. The system then proceeds with the constrained brute-force attack to recover the original plaintext and/or the encryption key (107).

FIG. 2 provides a conceptual illustration. A user (201) types on a keyboard (202). An Uninterruptible Power Supply (UPS) (203) sits nearby, containing a concealed audio bug (204). The bug records the sound of the typing (205). An analysis module (206) processes the sound and, instead of identifying the letters 'P-A-S-S-W-O-R-D', it extracts the metadata:

"Sequence Length: 8 characters" (207). This length information is the key output of the side-channel attack.

Claims

What is claimed is:

1. A computer-implemented method for a cryptographic side-channel attack, comprising the steps of:

    - capturing, via an audio sensor, the acoustic emanations of a user typing on a keyboard;
    - processing the captured audio with a processor to determine a metadata value corresponding to a number of keystrokes in a sequence;
    - providing said metadata value to a decryption system that is attempting to decrypt an encrypted data packet corresponding to the keystroke sequence; and
    - constraining a brute-force search for a decryption key by using the metadata value to eliminate from consideration all potential keys that would not result in a plaintext of a length equal to the number of keystrokes.

2. The method of claim 1, wherein the audio sensor is concealed within an Uninterruptible Power Supply (UPS).

3. The method of claim 1, wherein the metadata value is the exact character length of a password.

4. A system for assisting cryptographic analysis, comprising:

    - a covert audio recording device placed in proximity to a target keyboard;
    - an acoustic analysis module configured to receive audio from said device and to calculate the number of keystrokes in a detected typing sequence; and
    - an interface for transmitting the calculated number of keystrokes to a brute-force decryption engine, wherein the decryption engine is configured to use the number to limit its search space.

5. The system of claim 4, wherein the audio recording device is embedded within the housing of an Uninterruptible Power Supply (UPS).

6. The system of claim 4, wherein the acoustic analysis module is further configured to determine the rhythm and cadence of the typing sequence.

Drawings

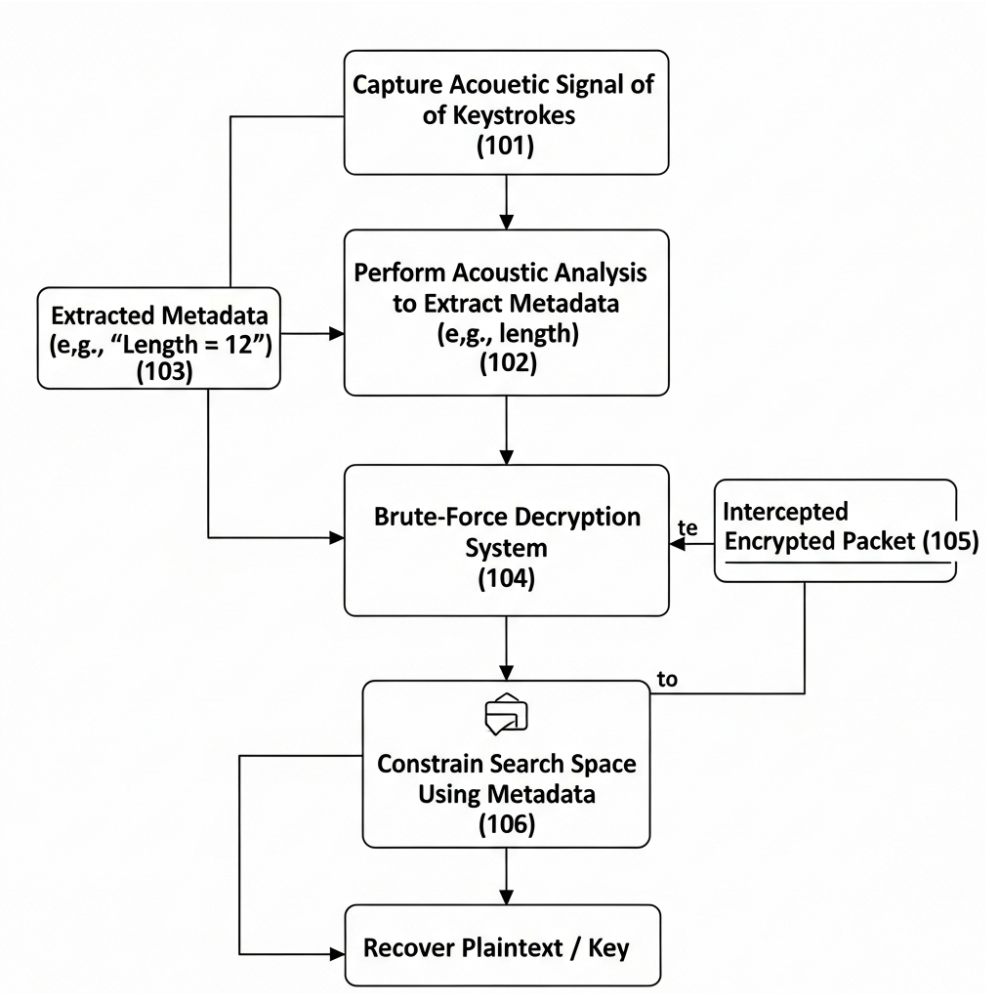


Figure 1: Flowchart of the acoustic side-channel attack method.

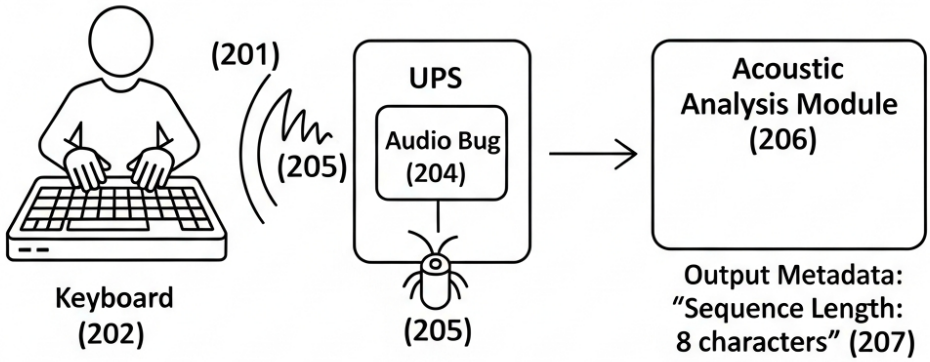


Figure 2: Conceptual diagram of metadata extraction.